

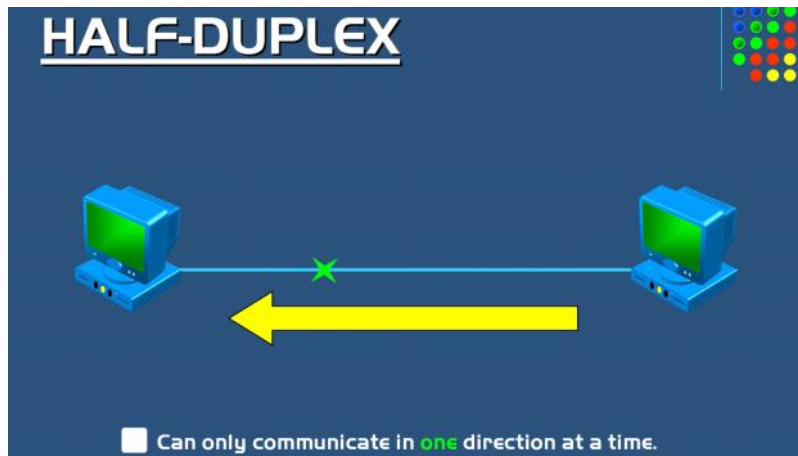
Half Duplex vs. Full Duplex Networking

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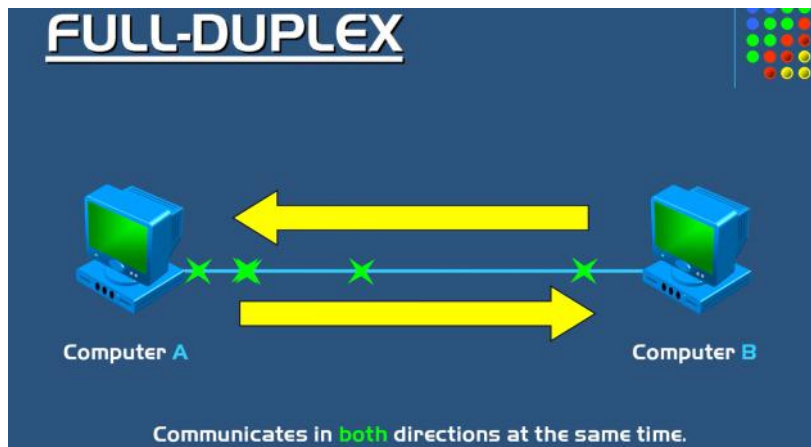
Half Duplex

- Data communication occurs in both directions, but only one direction is active at a time.
- Devices must take turns sending information; if one device is transmitting, the other must wait.
- A real-world analogy is a walkie-talkie, where you must finish speaking and release the button to hear the other person.
- Because data shares a single path, collisions can occur. Protocols like CSMA/CD (Carrier Sense Multiple Access with Collision Detection) are required to manage traffic and resolve these collisions.



Full Duplex

- Data communication occurs in both directions simultaneously.
- Devices can send and receive information at the same time, effectively doubling the available bandwidth compared to half duplex.
- A real-world analogy is a telephone call, where both parties can talk and listen at the same time without waiting.
- Since each direction has its own dedicated path, collisions are avoided, eliminating the need for collision management methods.
- Modern Usage



Most modern networks utilize full duplex technology. Half duplex is primarily found in obsolete or legacy hardware, such as network hubs, which are rarely used in current infrastructure.